



Classification of Hand-Grasp Movements of Stroke Patients using EEG Data

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Abstract— Electroencephalography (EEG) based Brain Controlled Prosthetics can potentially improve the lives of people with movement disorders, however, the successful classification of the brain thoughts into correct intended movement is still a challenge. In recent years, machine learning based methods, especially deep neural networks, have improved the pattern recognition and classification performance of computer vision systems. However, there is a need to evaluate the classical EEG signal processing algorithms against advanced machine learning based variants, specifically in the domain of Brain Computer Interfaces (BCI). This study aims to be a benchmark where we evaluate the performance of 5 popular motor imagery BCI pipelines and compare classical signal processing techniques (wavelet transform and power spectral density) with motor imagery specific algorithms (Common Spatial Patterns and Filter Bank Common Spatial Patterns (FBCSP)) and a state-of-the-art deep neural network based EEGNet algorithm. The experiments were performed on an open-source EEG dataset of hemiparetic stroke patients and both within subject and cross subject performance of the aforementioned algorithms was evaluated based on kappa scores. We empirically found that, for within subject classification, FBCSP method still is the gold-standard for motor imagery task with a mean kappa score of about 0.70 (84.8% accuracy) while for cross subject classification, due to the availability of a large amount of data, deep learning based EEGNet method outperformed all the other methods with a large margin and gave kappa value of 0.54 (77.0% accuracy). We believe these results would help BCI researchers to select a suitable BCI pipeline for their task that could help in the development of robot assisted therapies or as an interface for assistive devices.

Keywords—*Motor Imagery Classification, EEGNet, FBCSP, Brain Computer Interface (BCI), Assistive Devices*

I. INTRODUCTION

Brain Computer Interface (BCI) is the art of decoding the brain's electrical activity to produce control signals. A typical BCI system records brain signals, do some preprocessing, performs feature extraction, and classify the brain activity as one of the cognitive tasks like motor imagery, seizure prediction, etc. These types of systems may help people with neurological (Alzheimer, Epilepsy) and physical disabilities (Spinal Cord Injury, Paralysis) to interact with their environment in a better way and hence improve their quality of life.

Motor imagery (MI) BCI is a specific type of BCI in which a person imagines the movement of his arms/legs and his cognitive intention/activity is predicted by a BCI system. In most scenarios, EEG (electroencephalography) signals recorded over the motor cortex of the subject are passed to a BCI system that performs complex feature extraction and

classifies these signals to be one of the potential movement signals like moving the right hand. To move a step further, these signals may actuate an external control device like a prosthetic arm or wheelchair. These types of systems are particularly useful for people with motor disabilities like Spinal Cord Injury, Stroke patients, etc.

Motor Imagery BCIs typically monitor sensorimotor rhythms (oscillations) of the sensorimotor cortex; the brain areas responsible for the planning and control of voluntary motion. The brain's electrical activity can be measured in terms of the frequency of different oscillatory processes happening in the brain and these oscillations can be divided into 5 categories. They are Delta (< 4 Hz), Theta (4 -7 Hz), Alpha or Mu (8-12 Hz), Beta (12-30 Hz), and Gamma (30-100 Hz) oscillations. For non-invasive BCIs, It is observed that during motor imagination, alpha and beta bands go through event related synchronization/desynchronization (ERS/ERD) that can be used to detect the motor imagery task. In resting state, the brain oscillations are synchronized but when a person imagines a movement (say moving right hand) then the opposite hemisphere of the sensorimotor cortex (left in this case) goes through desynchronization (decrease in band power), and after the movement is complete the brain's electrical activity is again synchronized. So, these ERS/ERD patterns can help identify the type of movement from the EEG signal [1].

The task of hand grasp movement of a paralyzed person is a perfect use case of MI BCI and this study aims to set a benchmark based on existing top-performing BCI methods. For this purpose, we explored recent literature [1]–[5] and found that time-frequency based feature extraction methods (wavelet transform and power spectral density), spatial pattern based feature extraction methods (common spatial patterns and filter bank common spatial patterns), and deep learning based methods (EEGNet) are promising ones. Further details of these algorithms are provided in the methods section.

The key contributions of our work are:

- For benchmarking, we have picked 5 best performing feature extraction algorithms for motor imagery tasks across the domains of signal processing, classical machine learning, and deep learning.
- We have compared algorithms for both within subject and cross-subject performance.
- As the dataset is relatively new, this is the first benchmarking study on this dataset. Moreover, the dataset is recorded from actual hemiparetic stroke patients so our results are a true representation of actual neuro-prosthetics use case.

- The algorithms are implemented using Python (an open-source programming language). We have also open-sourced the code files for reproducibility. The code files with instructions can be found at <https://github.com/SulemanRasheed/EEG-HandGrasp-Classification>

II. MATERIALS AND METHODS

A. Dataset Description

The open-source dataset is provided by the organizers of Clinical BCI Challenge-WCCI 2020 and the description can be found at <https://github.com/5anirban9/Clinical-Brain-Computer-Interfaces-Challenge-WCCI-2020-Glasgow>. The dataset contains EEG data for 10 hemiparetic stroke patients with a hand functional disability. The patients were asked to attempt left and right-hand grasp movements while recording this data. The labels of original evaluation files weren't open-sourced so we used the training files only and performed cross-validation for evaluation.

All the training files contain balanced trials i.e. there are an equal number of trials for both classes in each file. The Experimental setup uses 512 Hz sampling frequency and contains 4096 samples in each trial confirming to overall 8 seconds of recording for each trial. The data were recorded from 12 channels that were arranged according to the 10-20 system. Labels are either '1' or '2', where '1' corresponds to the "right motor attempt" and 2 corresponds to the "left motor attempt". For data cleaning and noise removal, a bandpass filter of 0.1-100 Hz and a notch filter of 50 Hz was also applied.

B. Timing Details

All the trials last for 8 seconds with an inter-trial interval of 2-3 seconds. The modified timing diagram with $t=0$ at cue onset is shown in Fig 1. The first 3 seconds are the preparatory phase where a "Get Ready" message is displayed on the screen and an alarm beep is activated after 2 seconds. Then at the end of the 3rd second, the cue appears on the screen that lasts till the end of the trial and the subject imagines the displayed movements at this time. Moreover, the interval marked as 'sweet spot' for classification is the interval that we would use in this work for classification purposes. Please note that from now on, we would refer to the modified timing diagram whenever talking about timing details.

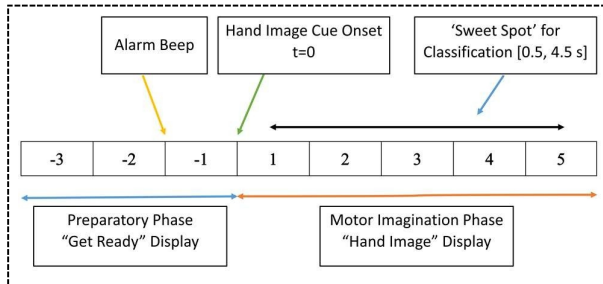


Fig 1. Modified Timing Diagram of Single EEG Trial

C. Preprocessing

We have used the MNE-Python package [6] to perform EEG preprocessing. The original EEG data was provided in .mat files so we first loaded it and converted it to MNE-Python compatible "epoch" datatype. Then we time locked the individual trials at cue onset and converted [0, 8 sec] trials to

[-3, 5 sec] trials as shown in Fig 1. Then we dropped the first two seconds because they were not a part of the motor imagery phase and we are left with [-1, 5 sec] trials. Then to avoid edge artifacts, we used a buffer time of about 500ms. This buffer time was found by a heuristic explained by M. Cohen [7] that the minimum buffer time should at least be 3 times the time period of the minimum frequency of interest. Here, the minimum frequency of interest is 7 Hz (as would be used in bandpass filter later) which corresponds to 428ms, so 0.5 sec seemed to be a good choice for this buffer time. The advantage of retaining one second before cue onset becomes clear now and this is also the reason we are excluding last 0.5 seconds from the 'sweet spot' for classification. Then, we specified International 10-20 system's montage with 12 given channel locations to plot topographical maps. Then we applied a 1 Hz high pass filter and applied baseline subtraction (also called baseline correction) with the interval of [-250, 0 ms] to avoid slow frequency drifts. Then to focus on just alpha and beta band activity, we applied a band pass filter of [7, 32 Hz].

D. Feature Extraction

In this study, we have compared 5 different EEG processing pipelines that combine feature extraction and classification. The details of each of the features extraction methods are as follows:

1) Moving Window CSP based Feature Extraction (CSP)

Common Spatial Patterns (CSP) [8] is one of the most widely used techniques in the MI BCI applications. It is a supervised feature extraction technique that uses training data to find maximum discriminating features for both classes such that variance for one class is maximum while variance is minimum for the other class [1]. In [9], authors have empirically shown that using overlapping segmented (moving) windows of the EEG with CSP gives much better results for motor imagery BCI task than applying CSP on the whole trial at once. We used a window length of 2.5 seconds in our analysis. For within subject analysis, we used 5 overlapping windows with a displacement of 375ms while for cross subject analysis we used 3 overlapping windows with a displacement of 250ms. The preprocessed EEG signal is divided into multiple segmented windows and the CSP algorithm is applied to each of them and classification is performed after feature fusion.

2) Power Spectral Densities (PSD) based Feature Extraction

The most intuitive way to detect event related oscillations (ERS/ERD) would be to observe the power spectrum activity. In this regard, Time-frequency techniques based PSD estimation algorithms like Short Time Fourier Transform (STFT) and multitapers are preferred over a simple Fast Fourier Transform (FFT) to capture non-stationary brain dynamics. In initial experiments, we used both Welch's method and multitaper method for PSD estimation and found that the multitaper method gives better results. So, we have used multitapers based PSD method for computation of PSD in the range of [8, 30 Hz].

3) Wavelet Transform (WPD+DWT) with CSP based Feature Extraction (WT-CSP)

In this method, we used wavelet transform based decomposition techniques [10] like Discrete Wavelet Transform (DWT) and Wavelet Packet Decomposition (WPD) to extract time-frequency features. DWT and WPD can be realized as a multistage subband filter where at each

stage we are applying a high pass and a low pass filter. The difference between DWT and WPD is that DWT further applies filters on only approximated coefficients while WPD applies filters to both detailed and approximated coefficients. So, WPD can give us more detailed insights at the cost of extra computation.

Two important hyperparameters of the wavelet transform are the choice of mother wavelet and the level of decomposition. The mother wavelet was chosen empirically for each subject on a set of wavelets including wavelets from the Daubechies family (db2, db4), Symlets (sym2, sym4), Haar (db1/Haar), and Coiflets (coif1, coif2, coif4). The original signal was sampled at 512 Hz so for DWT we used 5 decomposition levels and for WPD we used 3 decomposition levels that produced respectively 6 and 8 subbands. We empirically found that using all the coefficients improves the kappa scores so instead of using only alpha and beta band-specific coefficients, we have used all of them. So, overall we got 14 subbands and we calculated average power, mean and standard deviation in each subband as a feature. Then we applied the Common Spatial Patterns (CSP) algorithm on each subband and then performed features fusion (merged the features).

4) Filter Bank Common Spatial Patterns (FBCSP)

FBCSP [11] is an improved version of CSP algorithm. The CSP algorithm passes the EEG signal through a bandpass filter to find optimal discriminating features in the alpha and beta bands. Since these optimal subbands are different for different subjects so the performance of CSP based feature extractors is affected by the choice of frequency subband selection. To increase the robustness of the BCI system and to tune the algorithm for subject-specific frequency subband, the sub-band FBCSP algorithm first divides the preprocessed EEG signal to a series of band-pass filtered signals and then a CSP algorithm along with Linear Discriminant Analysis (LDA) classifier is applied on each of these bands individually and then score fusion and classification are performed.

5) EEGNet

EEGNet [12] is a compact deep Convolutional Neural Network (CNN) based deep learning classifier that was built to validate the idea that a CNN is BCI-paradigm agnostic and the same model can be used to learn different BCI tasks/paradigms. The authors utilized the idea of temporal, spatial, and separable convolution blocks to make a compact CNN optimized for and validated their neural network on 4

different BCI paradigms. The network is designed in such a way to have the least number of parameters so it can learn on a relatively small amount of training data and be suitable for realtime applications. This is achieved by utilizing the concept of depth-wise separable convolutions and pooling. The final layer is a fully-connected layer that performs classification. These references [5], [12], [13] can be consulted for further details.

For training, instead of using a classical single-trial training, we have used a cropped (moving window) training strategy where we divided the preprocessed EEG signal into overlapping windows and used each of them as a single example. This data augmentation helped increase the size of the training set that improved the performance of the deep learning model. We used a fixed window size of 2s and the value of inter-window displacement was 250ms for within subject decoding and we 1s for cross-subject decoding. So, we got 1360 training windows per subject for within subject decoding and 1920 total windows for cross subject decoding.

E. Classification

For Moving Window CSP, PSD, and WT-CSP methods, we have only chosen two linear classifiers for within subject classification

1. Linear Support Vector Machines (Linear SVM)
2. Linear Discriminant Analysis (LDA)

Whereas for cross subject classification we have used 3 extra classifiers

3. K Nearest Neighbors
4. Logistic Regression
5. Kernel SVM (with 'Rbf' kernel)

The reason for making these choices is that for within-subject classification, the training set size is only 80 samples so the complex models are more likely to overfit. So, for the first task, we have just used two simple linear classifiers while for the second task we have a lot of training data so, we have used 5 classifiers.

For reporting results we only used the best performing classifier. For Filter Bank Common Spatial Patterns (FBCSP), we only used LDA as a classifier while EEGNet itself do feature extraction and classification and doesn't need any other classifier.

F. EEG Processing Pipeline

The EEG processing pipeline is shown in Fig 2 and it works in the following fashion. The raw EEG signal is preprocessed and passed through one of the five feature

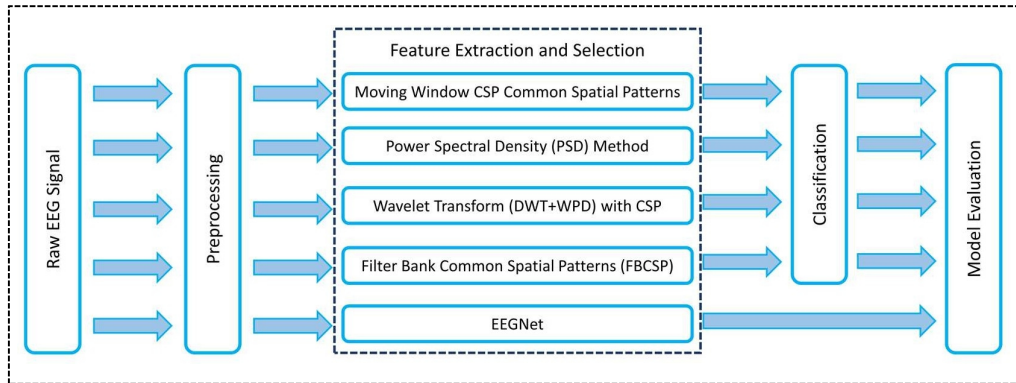


Fig 2. Proposed EEG Processing Pipeline

extraction methods. Then these features are standardized and then classified to give their predictions. These predictions are then compared with true labels and Cohen's Kappa Score [14] is used as an evaluation metric. Where possible, grid search was utilized to find the best performing models.

Cohen's kappa statistic is calculated as:

$$\kappa = \frac{p_0 - p_e}{1 - p_e} \quad (1)$$

Where p_0 is the relative observed probability (similar to accuracy), p_e is the hypothetical probability of chance agreement and κ is the Cohen's kappa statistic whose value is between -1 and 1. Kappa value of 0 suggests the chance agreement while the kappa value of 1 suggests perfect agreement [14]. For the case of balanced trials, $p_e = 0.5$ that leads us to a simplified relationship between observed probability (accuracy) and Cohen's kappa statistic given by

$$p_0 = 0.5(\kappa + 1) \quad (2)$$

We can use (2) to convert kappa scores to accuracy when the data is balanced. Our EEG dataset is balanced and we have also used stratification in cross-validation so we are comfortable in using (2) to interpret the results of our analysis.

For within subject classification, we utilized a grid search with shuffle split and stratification with 10 fold cross-validation to select the best hyperparameters. However, for EEGNet, we used hold-out cross validation with 75:25 due to high computational requirements for training. For the case of cross-subject classification, we used the leave one group (subject) out cross-validation for better generalization on unseen data.

G. Implementation Details

For implementation purposes, we used Python as the programming language. The Scikit-learn package [15] was used for the implementation of the machine learning algorithms while the MNE-Python package [6] was used for the EEG analysis. Moreover, the PyWavelets package [16] was used for the implementation of wavelet transforms i.e. DWT and WPD. The EEGNet implementation was taken from Braindecode [5] library. FBCSP implementation was taken from an open-source Github project: <https://github.com/jesus-333/FBCSP-Python>. The standard python packages for scientific computation, Numpy and Scipy, were also used frequently during analysis.

III. RESULTS AND DISCUSSION

For within subject classification, all methods reached about 80 percent accuracy or above. And the mean performance scores only differ by a slight margin across different algorithms. To be specific, the FBCSP method outperformed others and gave the highest mean accuracy of 84.8%. This can be attributed to the fact that for within subject decoding, the selection of multiple frequency subbands and then applying CSP on each of them gave an information rich feature set that was very helpful in decoding motor imagery task. It is interesting to note that moving window CSP and PSD based methods performed much better for the first 6 subjects. However, for subjects 7 and 8, their performance was really low as compared to other algorithms due to which the overall mean scores are less for these. This observation is also validated by a relatively higher standard deviation in the

results of these two algorithms. The WT-CSP method gave mean accuracy of 82.8 to secure second spot. The good performance of this algorithm can be explained by the fact that this method uses 14 wavelet coefficients and apply CSP individually on each one of them so the feature space is quite rich in discriminative power that's why we get very high kappa values for this method. Another important observation is that WT-CSP algorithm produced very stable results across the subjects and the standard deviation values are minimum. It is important to note that for within-subject classification, we have used [0.5, 4.5 sec] interval of the preprocessed EEG epochs. Moreover, while selecting frequency bands we explicitly used [8, 30 Hz] range for PSD, CSP, and FBCSP

Table 1. Within Subject Classification Results. Bold Scores represent the Best Performing Models. The last 2 rows show Mean and Standard Deviation of the Results across All Subjects.

Subject ID	EEG Processing Pipelines (Feature Extractors with Best Performing Classifiers)				
	CSP	PSD	WT-CSP	FBCSP	EEGNet
	Linear SVM	Linear SVM	LDA	LDA	EEGNet
P01	0.75	0.65	0.65	0.79	0.49
P02	0.85	0.75	0.775	0.79	0.70
P03	0.6	0.675	0.7	0.62	0.59
P04	0.575	0.775	0.6	0.64	0.38
P05	0.725	0.425	0.675	0.67	0.67
P06	0.525	0.825	0.575	0.78	0.71
P07	0.35	0.35	0.575	0.51	0.70
P08	0.35	0.4	0.7	0.77	0.77
Mean Kappa	0.59 ± 0.18	0.60 ± 0.19	0.66 ± 0.07	0.70 ± 0.10	0.63 ± 0.13
Mean Accuracy	79.5 ± 9.1	80.3 ± 9.4	82.8 ± 3.5	84.8 ± 5.2	81.3 ± 6.6

methods in all experiments.

For the case of cross subject classification, the kappa values are lower than that of within subject classification which shows that cross-subject classification is a more difficult task. We used a shorter time window of [1.5, 4.5 sec] for moving window CSP, PSD, and WT-CSP methods as they gave better results. The reason might be that the exact time delay between the cue onset and imagining the movement would be different across subjects, so allowing further delay after cue onset allows these cross-subject oscillations to better synchronize. Overall, deep learning based EEGNet is a clear winner and it outperformed all the other models with a large margin. EEGNet gave cross-subject decoding accuracy of 77% which is much higher than the other 4 methods. This large boost in performance is due to the fact that deep neural networks perform better with the availability of a large amount

Table II. Cross Subject Classification Results. Bold Scores represent the Best Performing Models.

Method	CSP	PSD	WT-CSP	FBCSP	EEGNet
Classifier	Linear SVM	Linear SVM	Kernel SVM	LDA	EEGNet
Best Kappa	0.416	0.434	0.353	0.40	0.54
Best Accuracy	70.8	71.7	67.6	70.0	77.0

of data and for cross-subject decoding task we had a relatively large amount of data so, it did great. While for within subject task, the data were relatively small so it couldn't outperform FBCSP in that task. It is interesting to note that the PSD method was runner up and all other methods that used CSP as a part of their pipeline couldn't perform well. This may be because these methods rely on the frequency subband

selection as both wavelets and CSP based methods build up their features on the frequency subband selection and for different subjects, the optimal subbands might be different that could be a cause of relatively lower kappa scores.

IV. CONCLUSION AND FUTURE WORK

In this study, we have successfully classified hand-grasp movements of stroke patients using 5 different methods based on time-frequency analysis, common spatial patterns, and deep learning and found that for within subject classification, the FBCSP method is still an optimal choice (about 85 % accuracy) due to its ability to select optimal frequency subbands for CSP filters. For cross-subject analysis, the deep learning based EEGNet model outperformed all the other methods (77% accuracy) due to the availability of large training data and can be effectively used for cross-subject decoding task.

In the future, we can further optimize the proposed EEG processing pipelines. In this study, we have used the moving window method strategy for CSP and EEGNet only, we can also use this with the other three schemes that might improve their results at the cost of extra computation. Moreover, we didn't use any artifact removal technique so one option might be to explore multiple artifact removal or signal denoising techniques like Independent Component Analysis (ICA), Multiscale PCA, etc. to improve the quality of the preprocessed EEG signal. Similarly, data augmentation techniques can be used to synthetically generate more training data that would improve the performance of deep learning based method for within-subject classification.

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